25900273 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBTG201,MG3DSEBBT202 - PLANT AND ANIMAL PHYSIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence

Answer any **10** questions

Each question carries **2** marks

1. Name two functions of plasma protein. [K] / [CO1]
2. Define reflex action. [K] / [CO1]
3. What is diffusion? [K] / [CO3]
4. Name two microbes that increase organic matter in soil. [K] / [CO4]
5. Name two compontents of circulatory system. [K] / [CO1]
6. Which hormone can replace vernalization? [K] / [CO3]
7. What is the primary function of respiratory pigments ? [K] / [CO1]
8. Name an example of a natural cytokinin. [K] / [CO3]
9. Name two examples of nitrogen - fixing bacteria. [K] / [CO4]
10. Define macronutrients. [K] / [CO3]

11. Who is the" Father of Modern Physiology "and write the meaning of "Physiology". [K] / [CO1]

12. Name the laboratory tests in the diagnosis of endocrine system disorders. [K] / [CO2]

[2x10 = 20]

Part B

Short Essay

Answer any **4** questions

Each question carries **5** marks

13. Explain the deficiency symptoms of any five macro elements. [U] / [CO3]

14. Leaves of Mimosa plant are sensitive to touch. Explain. [U] / [CO3]

15. Explain the traditional methods of crop improvement. [U] / [CO4]

16. Explain digestion and why it is important for survival? [U] / [CO1]

17. Discuss on the symptoms of lack of oxygen to the brain. [U] / [CO2]

18. Explain the different types of water absorption in plants. [U] / [CO3]

[5x4 = 20]

Part C

Essay type questions

Answer any **2** questions

Each question carries **15** marks

19. Discuss biofertilizers types,benefits and examples of microbes used in agriculture. [U] / [CO4]

20. Describe the process which is common in aerobic and anaerobic respiration. [U] / [CO3]

21. Describe common nutritional disorder,s causes,symptoms and preventive measures. [U] / [CO1]

22. Discuss on the common conditions,the disorders that affect lungs and its treatment? [U] / [CO2]

[15x2 = 30]